AZURE DATABRICKS ADVISORY

Cost Optimisation & Cost Management Strategy

A framework and roadmap for Microsoft Azure Databricks environments

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Date: 23 September 2026
Classification: Confidential — Client Use Only

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1. Executive Summary

Azure Databricks spend is driven by two independent meters — Databricks Units (DBUs) and the underlying Azure compute, storage and network resources the platform provisions. Left unmanaged, both grow in step with adoption, and organisations typically discover 20–40% of spend is avoidable once workloads, compute selection and governance are reviewed.

This document sets out a two-part strategy: a Cost Management framework for visibility, governance and accountability, and a Cost Optimisation programme covering compute, storage, network and commercial levers. It concludes with a phased implementation roadmap and the outcomes each phase is expected to deliver.

> **Key principle**
> Optimise before you commit. Reserved capacity and DBCU pre-purchases lock in whatever baseline is in place at the time of purchase — waste included. Governance and quick-win optimisation should run first; commercial commitments follow once the baseline is stable.

2. Understanding the Azure Databricks Cost Model

Every Databricks invoice is the sum of two separate charges that must be managed together:

Cost component	Billed by	What drives it
Databricks Units (DBUs)	Databricks, via the Azure Marketplace	Compute type (Jobs, All-Purpose, SQL Warehouse, Serverless), runtime used, and hours running
Azure infrastructure	Microsoft Azure directly	VM family and size, managed disks, storage (ADLS), networking, NAT Gateway and egress

Azure infrastructure is frequently 40–60% of total Databricks-related spend, yet most cost reviews focus only on the DBU line because that is what appears on the Databricks usage dashboard. A complete view requires joining Databricks usage data with the Azure Cost Management export for the same subscriptions.

Serverless compute (SQL, jobs and notebooks) collapses both meters into a single DBU-based rate. This simplifies attribution but shifts the optimisation levers from infrastructure tuning toward workload efficiency and warehouse sizing, which the levers in Section 4 account for.

3. Cost Management Framework

Cost management is about visibility and control — knowing where spend originates, attributing it correctly, and preventing uncontrolled growth. It is the foundation the optimisation programme in Section 4 is built on.

3.1 Tagging Standard

A consistent tagging standard is the single highest-leverage governance control, because it lets every downstream report attribute cost to a team, project or environment automatically.

- **CostCenter** — the budget the workload draws against
- **Project** — the initiative or product the workload supports
- **Environment** — Production, Staging or Development
- **Owner** — the individual or team accountable for the workload

Tags should be enforced through cluster policies rather than requested by convention, so they propagate consistently to both the Databricks usage records and the underlying Azure VMs and disks.

3.2 Visibility: System Tables and Dashboards

Databricks system tables provide native, queryable billing data that can be joined with Azure Cost Management exports for a single reconciled view.

- Build a cost dashboard on system.billing.usage joined with system.billing.list_prices, broken down by workspace, SKU, job, warehouse and tag.
- Reconcile the Databricks-reported spend against the Azure Cost Management export for the same subscription every month, to catch infrastructure costs the DBU view misses.
- Track unit economics alongside raw spend: cost per pipeline run, cost per TB processed, and cost per dashboard query, so cost trends can be separated from usage growth.

3.3 Budgets, Alerts and Policy Controls

- Set Azure Cost Management budgets and anomaly alerts per subscription and resource group.
- Use Databricks serverless budget policies to attribute and cap serverless usage, which otherwise falls outside standard VM-based controls.
- **Compute policies:** restrict cluster creation so only administrators can launch unrestricted clusters; everyone else works within policies that cap maximum workers, restrict allowed VM families, mandate tags and force auto-termination.

3.4 Chargeback and Ownership

- Publish a monthly cost report per business unit or project, showing spend against budget and against the prior month.
- Assign a named FinOps owner accountable for the Databricks cost programme, and run a monthly cost review with representation from engineering and finance.

Visibility and accountability alone — before any technical optimisation — typically produce a 10–20% reduction, simply because cost becomes something individual teams can see and are asked to explain.

4. Cost Optimisation Levers

With governance in place, the following levers reduce the underlying spend. They are grouped by compute, data and storage, network, and commercial terms, roughly in order of typical impact.

4.1 Compute — the largest lever

- **Match compute type to workload:** run production pipelines on Jobs Compute rather than All-Purpose clusters — the DBU rate is roughly half. Reserve All-Purpose clusters for interactive development, and use SQL Warehouses or serverless for BI and dashboards.
- **Autoscaling and auto-termination:** enable both on every cluster; set interactive clusters to auto-terminate after 15–30 minutes of inactivity. Use cluster pools for jobs that start frequently, to cut cold-start time and cost.
- **Spot VMs:** use spot instances for worker nodes on fault-tolerant batch jobs, keeping the driver on-demand with automatic fallback to on-demand workers if spot capacity is unavailable. Typical savings are 60–80% on the affected worker VM cost.
- **Right-sizing:** review cluster utilisation metrics regularly and resize or reduce node counts on consistently under-used clusters. Choose VM families to match the workload — memory-optimised for joins and shuffles, compute-optimised for ETL-heavy stages.
- **Photon:** Photon carries a higher DBU rate per hour but frequently completes SQL and Delta-heavy jobs 2–3x faster, making it net cheaper. Benchmark representative workloads before enabling it broadly.
- **Shared job clusters:** in multi-task workflows, share a single job cluster across tasks rather than spinning up a new cluster per task, to avoid repeated startup cost.
- **SQL Warehouse sizing:** prefer serverless SQL Warehouses with aggressive auto-stop (5–10 minutes), right-sized T-shirt sizing, and a capped maximum number of clusters per warehouse.

4.2 Data and Storage

- **Delta table hygiene:** enable predictive optimisation, or schedule regular OPTIMIZE jobs; use liquid clustering instead of deep, static partitioning; run VACUUM on a sensible retention schedule to remove stale files.
- **ADLS lifecycle policies:** move bronze and raw data to Cool or Archive storage tiers on a defined schedule, and delete stale checkpoints and temporary job output automatically.
- **Incremental processing:** use Auto Loader and incremental patterns (or Delta Live Tables) instead of full-table reloads, and use MERGE rather than overwrite where only a subset of data has changed.

4.3 Network

- Keep the Databricks workspace, storage account and downstream consumers in the same Azure region to avoid cross-region transfer charges.
- On VNet-injected workspaces, monitor NAT Gateway and egress charges specifically — these are easy to overlook and can grow quickly with high-throughput jobs.

- Use Private Endpoints deliberately rather than by default, weighing the security benefit against the added networking cost.

4.4 Commercial Terms

- **Databricks Commit Units (DBCU):** pre-purchase DBUs through Azure, on a 1- or 3-year term, at a discount to pay-as-you-go rates — once usage patterns are stable enough to size the commitment accurately.
- **Azure Reserved VM Instances or Savings Plan:** cover the always-on baseline workload (steady-state jobs, always-on SQL Warehouses) with Reserved Instances or a Savings Plan for compute.
- **MACC alignment:** Databricks spend purchased through the Azure Marketplace counts toward a Microsoft Customer Agreement consumption commitment, and Azure Hybrid Benefit can reduce licensing costs on eligible VM families.

5. Implementation Roadmap

The programme is sequenced so that governance and quick wins are in place before any commercial commitment is made, and so that engineering-heavy optimisation work happens against a stable, well-understood baseline.

Phase	Timeline	Focus	Key activities
1. Visibility	Weeks 0–4	See the spend	Enable system tables; roll out the tagging standard; build the reconciled cost dashboard; set budgets and anomaly alerts
2. Quick wins	Weeks 4–8	Stop the leaks	Apply compute policies; enforce auto-termination; move production jobs to Jobs Compute; enable spot instances and warehouse auto-stop
3. Engineering optimisation	Months 2–4	Tune the workloads	Right-size clusters; benchmark and roll out Photon; adopt Delta optimisation and incremental pipelines; apply storage lifecycle policies
4. Commercial & continuous	Month 4 onward	Lock in the gains	Purchase DBCU and VM reservations against the stabilised baseline; establish chargeback; run monthly FinOps reviews; enforce policy-as-code

Sequencing rule

Phases 1 and 2 typically deliver 20–35% savings on their own within the first eight weeks, using governance and configuration changes alone — no code changes and no commercial commitment required. Phases 3 and 4 build on that stabilised baseline.

6. Expected Outcomes

Savings vary by starting maturity, but the following ranges are typical for each lever when applied to an unoptimised Azure Databricks estate:

Lever	Typical savings	Effort
Visibility and chargeback (governance only)	10–20%	Low
Compute policies + auto-termination	10–15%	Low
Jobs Compute vs. All-Purpose for production	Up to ~50% on affected DBUs	Low–Medium
Spot instances for fault-tolerant batch workers	60–80% on affected VM cost	Medium
Right-sizing and Photon adoption	10–25%	Medium
Delta optimisation and storage lifecycle	5–15% on storage and I/O-bound jobs	Medium
DBCU and Reserved Instance commitments	Additional 15–30% on committed baseline	Low (post-stabilisation)

These figures are illustrative benchmarks, not guarantees — the roadmap in Section 5 includes a discovery phase specifically to establish the current baseline and size each opportunity against this client's actual workload mix before committing to targets.

7. Next Steps

1. Confirm access to the relevant Databricks workspaces and Azure subscriptions to begin the Phase 1 discovery and dashboard build.
2. Agree the tagging standard and cluster policy set with engineering leadership.
3. Schedule the first monthly FinOps review and confirm the accountable owner on the client side.
4. Review this document together and agree the target date for the Phase 1 visibility milestone.

Prepared by Uptut — Azure Databricks training and advisory.